

Bulk RNA-seq Differential Gene Expression Analysis of Glioblastoma

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Introduction

Glioblastoma (GBM) is the most aggressive primary malignant brain tumor in adults. Despite advances in surgery, radiotherapy, and chemotherapy, the prognosis remains poor due to rapid tumor progression and extensive molecular heterogeneity. Bulk RNA sequencing (RNA-seq) enables comprehensive analysis of gene expression changes occurring during tumor development and provides valuable insights into the molecular mechanisms underlying glioblastoma progression. The objective of this project was to identify differentially expressed genes between healthy brain tissue and glioblastoma samples and to investigate enriched biological processes associated with tumor biology.

Biological Question

Which genes and biological processes are significantly altered in glioblastoma compared with healthy brain tissue?

Dataset

The publicly available RNA-seq dataset GSE147352 was obtained from the Gene Expression Omnibus (GEO).

Dataset information

- Database: Gene Expression Omnibus (GEO)
- Accession: GSE147352
- Organism: Homo sapiens
- Technology: Bulk RNA sequencing

The analysis included healthy brain tissue and glioblastoma samples. Lower-grade glioma samples were excluded from downstream analysis to allow a direct comparison between healthy tissue and GBM.

Methods

The complete bioinformatics workflow was implemented in RStudio.

The analysis consisted of the following steps:

1. Import of the raw count matrix.
2. Download of sample metadata using GEOquery.
3. Preparation of sample metadata.
4. Differential gene expression analysis using DESeq2.
5. Gene annotation using the org.Hs.eg.db annotation database.
6. Identification of significantly differentially expressed genes.
7. Gene Ontology (GO) Biological Process enrichment analysis.
8. Visualization of results using:
 - MA Plot
 - Volcano Plot
 - GO Dotplot
 - GO Barplot
 - Principal Component Analysis (PCA)
 - Heatmap of the top 30 differentially expressed genes

Genes were considered significantly differentially expressed when:

- adjusted p-value (padj) < 0.05
- $|\log_2 \text{Fold Change}| > 1$

Software and Packages

The analysis was performed using:

- RStudio
- GEOquery
- DESeq2
- AnnotationDbi
- org.Hs.eg.db
- clusterProfiler
- enrichplot
- ggplot2
- ggrepel

- heatmap

Results

Differential Gene Expression

DESeq2 identified numerous genes significantly dysregulated between healthy brain tissue and glioblastoma samples. Several well-known glioblastoma-associated genes, including TOP2A, MKI67, NES, TNC, and F2R, showed strong differential expression.

MA Plot

The MA plot illustrates global gene expression changes after normalization and highlights genes exhibiting statistically significant differential expression.

Volcano Plot

The volcano plot demonstrates a clear separation between significantly upregulated and downregulated genes. Genes with the largest expression changes were highlighted to facilitate biological interpretation.

Gene Ontology Enrichment

Functional enrichment analysis revealed significant overrepresentation of biological processes associated with:

- immune response
- leukocyte migration
- chemotaxis
- exocytosis
- regulation of membrane potential

These findings suggest extensive remodeling of the tumor microenvironment and activation of immune-related signaling pathways in glioblastoma.

Principal Component Analysis

Principal Component Analysis demonstrated a clear separation between healthy brain tissue and glioblastoma samples. This indicates that the overall transcriptional profiles differ substantially between the two biological conditions.

Heatmap

The heatmap of the top 30 differentially expressed genes showed distinct clustering of healthy and glioblastoma samples. Genes involved in cell proliferation and extracellular matrix remodeling exhibited characteristic expression patterns across both groups.

Discussion

The differential expression analysis successfully identified molecular alterations associated with glioblastoma. The enrichment of immune-related biological processes is consistent with previous studies reporting extensive immune infiltration and inflammatory signaling within the glioblastoma microenvironment. Genes such as TOP2A and MKI67 are well-established markers of cellular proliferation, while NES and TNC have been associated with glioblastoma stem cell biology and tumor invasiveness. Overall, the obtained results demonstrate that the implemented bioinformatics workflow successfully captures biologically meaningful transcriptional changes characteristic of glioblastoma.

Conclusions

This project demonstrates a complete Bulk RNA-seq differential gene expression workflow implemented in R. The analysis identified numerous significantly dysregulated genes and revealed biological processes involved in glioblastoma progression. The workflow integrates differential expression analysis, functional enrichment analysis, and multiple visualization techniques, providing a reproducible framework for transcriptomic analysis of publicly available RNA-seq datasets.

Repository Structure

```
Bulk-RNAseq-Glioblastoma-DESeq2
|
├── README.md
├── scripts
│   └── 01_bulk_rnaseq_analysis.R
|
├── data
│   └── counts
|
├── figures
│   ├── MA_plot.png
│   ├── volcano_plot.png
│   ├── G0_dotplot.png
│   ├── G0_barplot.png
│   ├── PCA_plot.png
│   └── Heatmap_top30_genes.png
|
└── results
    ├── DESeq2_results_GBM_vs_Healthy.csv
    ├── significant_genes_GBM_vs_Healthy.csv
    └── G0_BP_results.csv
```

References

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